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#EL OTMAI Youssef
#worked with CHARKI Omar

Ring.<x> = QQ[]

a = x^2 + x + 1

b =( (3*(x^4) - 3*x + 9 )^2 ) * ( x - 5*(x^2) )

h = a * b
print("h = ", h, "\n")

k = a - b
print("k = ", k, "\n")

print("degree of h = ", h.degree(), "\n")
print("degree of k = ", k.degree(), "\n")
print("leading coefficient of h = ", h.lc(), "\n")
print("leading coefficient of k = ", k.lc(), "\n")

def DIVPOL(f,g):
    if(g==0) :
        return ('impossible', 'impossible')

    if(f==0) :
        return (0,g)

    R = Ring(f)
    G = Ring(g)
    Q = 0

    while (R!= 0) & (R.degree() >= G.degree()) :
        y = (R.lc()/G.lc())*x^(R.degree()-G.degree())
        R = R - y*G
        Q = Q + y

    return (Q,R)

f = 5*x^6 + 15*x^5 + 16*x^4 + 19*x^3 + 22*x^2 + 23*x + 14
g = x^2 + 3*x + 3

(q,r) = DIVPOL(f,g)

print(f, "divided by", g , "is \n",  q, "and the remainder is ",r, "\n")

(q,r) = DIVPOL(f,0)

print(f, "divided by", 0 , "is \n",  q, "and the remainder is ",r, "\n")

(q,r) = DIVPOL(f,2)

print(f, "divided by", 2 , "is \n",  q, "and the remainder is ",r, "\n")

(q,r) = DIVPOL(0,g)

print(0, "divided by", g , "is \n",  q, "and the remainder is ",r, "\n")

m= 3*x^3+ 5*x^2 + 4*x
n = x - 1

(q,r) = DIVPOL(m,n)

print(m, "divided by", n , "is \n",  q, "and the remainder is ",r, "\n")

def EXTEUC(a,b):
    A = Ring(a)
    B = Ring(b)
    S = 1
    T = 0
    U = 0
    V = 1

    while B != 0 :
        (QUOT, REM) = DIVPOL(A, B)
        A = B
        B = REM
        S1 = S
        T1 = T
        S = U
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        T = V
        U = S1 - QUOT*U
        V = T1 - QUOT*V

    c = A.lc()
    A = (1/c) * A
    S = (1/c) * S
    T = (1/c) * T

    return (A, S, T)

print("gcd ", EXTEUC(x^2 -1, x^2 +2*x +1)[0] , "= (x^2 -1) * ", EXTEUC(x^2 -1, x^2 +2*x +1)[1], " + (x^2 +2*x

Out :
k =  45*x^10 - 9*x^9 - 90*x^7 + 288*x^6 - 54*x^5 + 45*x^4 - 279*x^3 + 460*x^2 - 80*x + 1

degree of h =  12

degree of k =  10

leading coefficient of h =  -45

leading coefficient of k =  45

5*x^6 + 15*x^5 + 16*x^4 + 19*x^3 + 22*x^2 + 23*x + 14 divided by x^2 + 3*x + 3 is
 5*x^4 + x^2 + 16*x - 29 and the remainder is  62*x + 101

5*x^6 + 15*x^5 + 16*x^4 + 19*x^3 + 22*x^2 + 23*x + 14 divided by 0 is
 impossible and the remainder is  impossible

5*x^6 + 15*x^5 + 16*x^4 + 19*x^3 + 22*x^2 + 23*x + 14 divided by 2 is
 5/2*x^6 + 15/2*x^5 + 8*x^4 + 19/2*x^3 + 11*x^2 + 23/2*x + 7 and the remainder is  0

0 divided by x^2 + 3*x + 3 is
 0 and the remainder is  x^2 + 3*x + 3

3*x^3 + 5*x^2 + 4*x divided by x - 1 is
 3*x^2 + 8*x + 12 and the remainder is  12

gcd  x + 1 = (x^2 -1) *  -1/2  + (x^2 +2*x +1) *  1/2
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